

Vision2MeshAI

2D Image / Blueprint / Drawing → 3D Model Conversion System

1. Project Overview

Vision2MeshAI is an AI-based 2D-to-3D conversion project. The goal is to take visual information from 2D views such as blueprints, toy-model drawings, and suitable object illustrations and reconstruct a 3D representation. The current research prototype focuses on multi-view image understanding and voxel-based 3D reconstruction.

2. The Problem

Creating a 3D model manually from 2D drawings or reference images can take significant time and requires 3D modelling knowledge. A 2D image contains projected visual information, while a 3D model needs information about depth, shape, volume and spatial structure.

Core problem: How can an AI system understand multiple 2D views of an object and reconstruct its three-dimensional shape automatically?

3. Project Idea

The project uses multiple views of the same object—front, side and top—to give the AI complementary information. A deep-learning model extracts visual features from the three views, combines those features, and predicts a 3D voxel representation of the object.

5. Why Multiple Views?

A single 2D view can hide important geometry. The front view provides width and height information, the side view provides another depth-related constraint, and the top view provides another spatial constraint. Combining the three views gives the network more information for reconstructing shape.

7. Dataset

The current supervised prototype uses the ModelNet10 mesh collection. The project contains **4,899 real meshes** used to generate paired training data. For every object, the pipeline produces front, side and top rendered images and a corresponding voxel target.

The current 64^3 supervised dataset contains **4,899 voxel targets**. The image views are resized to 224×224 during training.

8. 32^3 Baseline

A 32^3 reconstruction model was trained first as a baseline. Its final evaluation produced approximately **0.7455 mean Dice** and **0.6171 mean IoU**. Visual inspection showed that the model learned global object shape, while small structures, gaps and fine geometric details were often lost at 32^3 resolution.

9. 64^3 Reconstruction Upgrade

The project was extended to 64^3 voxel reconstruction to provide a finer 3D representation. The 64^3 dataset was generated and verified with all 4,899 targets available. Because the system is being trained on CPU, the model and training pipeline were optimized. The verified configuration uses 224×224 image inputs, batch size 1, eight CPU threads and RAM-cached data.

12. Project Workflow

2D Views → Image Pre-processing → CNN Feature Extraction → Multi-View Fusion → 3D Voxel Decoder → 64³ Voxel Model → Mesh Reconstruction → 3D Visualization / Export

16. Simple Explanation for Presentation

Problem: A 2D drawing gives us only flat visual information, but creating a 3D model manually takes time and modelling knowledge.

Idea: Give the AI multiple views of the same object so it can understand the object's shape from different directions.

Solution: Our neural network extracts features from the front, side and top views, combines them, and predicts a 3D voxel representation. That representation can then be converted into a 3D mesh.

Result: The prototype has a complete 4,899-sample supervised dataset, a working 32³ baseline, and a verified 64³ reconstruction pipeline. The 64³ model has successfully completed a full-dataset 1-epoch training run.

4. Proposed Solution

Stage	What happens
1. Input	Front, side and top 2D views are provided.
2. Pre-processing	Views are resized and normalized consistently.
3. Feature extraction	A CNN-based image encoder extracts visual features from each view.
4. Multi-view fusion	Features from the three views are combined into a shared representation.
5. 3D reconstruction	A 3D decoder converts the fused representation into a voxel volume.
6. 3D output	The predicted voxel representation can be converted into a visible 3D mesh.

6. Main AI Architecture

Component	Purpose
Image Encoder	Extracts visual features independently from front, side and top images.
Multi-View Fusion	Combines the three feature maps into a shared representation.
Voxel Decoder	Transforms the fused representation into a 3D voxel volume.
Loss Function	Binary Cross Entropy + Dice loss trains voxel occupancy predictions against the target.

10. Current 64³ Training Status

Item	Current result
Complete samples	4,899 / 4,899
Training split	3,429
Validation split	734
Test split	736
Input image size	224 × 224
Voxel resolution	64 × 64 × 64
Batch size	1
CPU threads	8
Model parameters	1,145,802
Verification run	1 epoch completed successfully

Training loss	1.025812
Validation loss	0.893708
Checkpoint	C:\Users\Sowmya\Desktop\Vision2MeshAI\checkpoints_64\full_64_best_15epoch.pth

11. Technology Stack

Area	Technology / Tool
Programming	Python
Deep Learning	PyTorch
Image processing	Pillow, NumPy
3D processing	Trimesh / voxel processing pipeline
Backend prototype	FastAPI
Frontend prototype	Vite + React
Data	ModelNet10-derived supervised dataset

13. Expected Use Cases

- Converting technical or conceptual 2D references into an initial 3D representation.
- Generating a starting point for 3D modelling workflows.
- Supporting rapid visualization of objects from multiple 2D views.
- Providing a foundation for future 3D-printing-oriented reconstruction workflows.
- Reducing manual modelling effort for suitable object categories.

14. Current Limitations

- Current supervised data is based on ModelNet10 object categories.
- Voxel reconstruction is resolution-limited; fine details may still be lost.
- The 64³ model has completed a successful 1-epoch full-dataset run; the 15-epoch run is the next training stage.
- Quality depends on information available in the input views.
- Production use would require testing on real-world drawings and blueprints outside the training distribution.

15. Future Development

- Complete and evaluate the 15-epoch 64³ training.
- Measure 64³ test-set Dice and IoU.
- Compare predicted 3D meshes against ground truth visually.
- Improve fine-detail reconstruction.
- Expand toward real drawings and blueprint inputs.
- Add a user-facing upload interface and downloadable 3D output.
- Investigate higher-resolution or mesh-native reconstruction methods.